

Pedro Enrique Ramirez Lopez

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Admissions Committee, Online Master of Science in Computer Science (OMSCS)

I am pleased to submit my application to the Online Master of Science in Computer Science (OMSCS) program at the Georgia Institute of Technology. With more than two decades of experience in information technology, software development, data analytics, automation, and digital transformation, I am seeking to further strengthen my computer science foundation while expanding my expertise in artificial intelligence and advanced computing technologies.

Throughout my career, I have worked at the intersection of business and technology, leading initiatives involving software development, data engineering, analytics, robotic process automation, process mining, conversational AI, machine learning, and generative AI. In my current role leading Digital Business Solutions initiatives across the Americas, I collaborate with cross-functional teams to identify, design, and implement technology solutions that improve operational efficiency, enhance decision-making, and create measurable business value.

Georgia Tech's OMSCS program is particularly appealing because of its academic rigor, flexibility for working professionals, and strong curriculum across computer science disciplines. I am especially interested in advancing my knowledge of artificial intelligence, machine learning, computing systems, software architecture, and data-driven problem solving. The opportunity to learn from Georgia Tech faculty while studying topics grounded in both theoretical foundations and practical applications aligns directly with my professional goals.

In addition to my professional experience, I bring a strong academic background in computer science, project management, and business administration, complemented by graduate-level and professional training in artificial intelligence and machine learning. Over the years, I have learned that successful technology solutions require not only innovative algorithms but also scalable architectures, sound engineering practices, effective governance, cybersecurity awareness, and organizational adoption.

My long-term objective is to become a stronger technical leader capable of designing and delivering intelligent systems that are scalable, reliable, secure, and ethically responsible. I believe the OMSCS program will provide the advanced computer science knowledge necessary to complement my practical experience and prepare me for increasingly complex challenges in AI-driven digital transformation.

Thank you for considering my application. I look forward to the opportunity to contribute to and learn from the Georgia Tech OMSCS program as I continue my professional and academic growth.

Sincerely,
Pedro Ramirez

Academic and Technical Profile

Artificial Intelligence and Data: Machine Learning and Applied AI Generative AI and Large Language Models Conversational AI and Document Understanding Data Analytics, Data Engineering, and Visualization Process Mining and Process Intelligence Automation and RPA Solution Design	Computer Science and Systems: Software Development and Systems Engineering Database Management and Relational Data Models Business Intelligence and Data Warehousing Secure Software Development and Cybersecurity System Integration, Migration, and IT Governance Risk Evaluation and IT Auditing	Academic and Professional Strengths: Analytical Problem Solving Research-Oriented Technology Evaluation Technical Documentation and Executive Communication Project Management and Cross-Functional Collaboration Responsible AI, Governance, Compliance, and Ethics
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Experience

Associate Director - Digital Business Solutions Americas
July 2021 – Present

Academic-Relevant Experience and Technical Contributions:

- Annual Budget Planning for the DBS team: Collaborated in the preparation and management of the annual budget for the Digital Business Solutions (DBS) area, covering licenses, projects, and operational initiatives across North and South America. Delivered budgetary forecasts and ensured optimization of financial resources to support strategic IT initiatives while aligning spending with organizational priorities.
- Vendor and Contract Management: Managed IT service contracts exceeding \$1M with global technology providers to ensure cost savings and high-quality service delivery. Oversaw software license agreements and renewals for critical digital tools such as Process Mining and UiPath. Partnered with external vendors and technology teams to deliver innovative solutions, such as Generative AI integration, RPA implementations, and AI-powered process optimization, while maintaining adherence to organizational standards and timelines. Established measurable service delivery metrics (SLAs) and monitored vendor performance to drive efficiency across digital transformation initiatives.
- Established Process Mining Excellence: Directed implementation of multiple Process Mining projects across the USA (USIG, Corporate Tax, Corporate Procurement) and North Latin America, delivering actionable insights and measurable business benefits. Strategically evaluated and tracked cost-saving improvements, enhancing operational efficiency and aligning initiatives with broader business objectives.
- Automation Innovator: Identified and deployed Robotic Process Automation (RPA) use cases regionally across the Americas (e.g., USIG, Corporate Tax, Lincare) to achieve process optimization and reduce costs. Designed and implemented scalable RPAs that directly contributed to productivity growth and ROI.
- Pioneering Artificial Intelligence Integration: Leveraged AI technologies, including Machine Learning, Conversational AI, and Document Understanding, to identify reworks, workflow inefficiencies, business anomalies, and process gaps. Successfully implemented Generative AI (LLM) to streamline operations, generate content, and enable efficient query resolution for business stakeholders.
- Strategic Technology Leadership: Defined the two-year strategic roadmap for the integration of Process Mining, RPAs, Conversational AI, and other digital technologies based on evolving business

demands. Orchestrated initiatives that aligned IT systems and tools with short- and long-term business strategies.

- Global Team Collaboration: Led cross-functional development teams in Germany, Slovakia, India, the Philippines, and Hungary to drive project execution and foster collaboration across diverse geographies. Delivered consistent business outcomes while navigating cultural and operational complexities.

Senior Manager IT Audits, Praxair Inc. / Linde plc. (Praxair – Linde merger in 2018),
Aug 2015 – July 2021

Academic-Relevant Experience and Analytical Contributions:

- Global IT Audit Leadership: Directed IT audits across North America, Europe, Asia, and South America, evaluating key systems, processes, and controls in subsidiaries of Linde plc. Key focus areas included Cybersecurity, SOX Compliance, Application Development, Service Management, and Project Management. Delivered actionable recommendations based on leading IT governance frameworks such as COBIT and PMP, strengthening enterprise-level IT operations and driving continuous improvement.
- Operational Risk and Compliance Management: Conducted thorough evaluations of plant control systems for US businesses and provided risk assessment insights to improve system reliability and operational compliance.
- Data Analytics Program Implementation: Designed, developed, and maintained a comprehensive Data Analytics Program for IT Internal Audit, enabling enhanced monitoring of financial and operational cycles including Procure-to-Pay (PTP), Record-to-Report (RTR), and Order-to-Cash (OTC) processes. Automated segregation of duties within platforms such as JDE and Active Directory, driving data governance improvements backed by dashboards for real-time insights.
- Performance Measurement Innovation: Initiated the creation of Key Performance Indicators (KPIs) for the Internal Audit department, enabling performance tracking at the auditor, regional, and global levels. Tableau dashboards visualized audit efficiency, effectiveness, and agility, fostering transparency and driving accountability across stakeholders.
- Technology Integration Expertise: Orchestrated technical solutions to optimize audits and analytics processes, integrating data extraction, transformation, and visualization across multiple platforms to streamline decision-making and strengthen IT governance.

Highlights:

- Promoted from IT Auditor (2015) to IT Audit Manager (2016) and subsequently to Senior Manager (2018), reflecting proven leadership and technical proficiency.
- Expanded responsibilities following the merger of Praxair Inc. and Linde plc (2019), delivering audit coverage and leadership in a significantly broader operational structure.

IT Manager, Applications Development / Business Intelligence, Praxair Mexico
Aug 2008 – Jul 2015

Academic-Relevant Experience and Systems Contributions:

- Budget Management and Execution: Prepared and administered IT budgets for Mexico's Central Region, ensuring alignment of project execution with financial plans and delivering initiatives on time and within scope.
- Vendor and Contract Management: Negotiated and oversaw contracts with technology vendors and software providers, ensuring cost-effective solutions and alignment with Praxair Mexico's IT strategy. Managed licensing agreements and renewals for enterprise software ensuring compliance and uninterrupted service delivery across IT operations.

- Enterprise Application Leadership: Led project management for key IT initiatives, including the implementation and maintenance of ERP modules using JDE Edwards, streamlining operational workflows for gases and cylinder businesses in Mexico. Successfully integrated legacy applications and systems into JDE Edwards via API interfaces, showcasing expertise in facilitating enterprise-wide system transitions.
- Business Intelligence Solutions: Spearheaded the design and implementation of Cognos-based Business Intelligence (BI) solutions, crafting ETL processes and reporting data marts for accounting, logistics, and sales. Established Distribution KPIs leveraging TM1 (Cognos) for performance oversight and operational improvement.
- Mobile Technology Innovation: Developed critical mobile solutions for Android and iOS platforms using IDE tools, advancing business productivity through user-friendly mobile interfaces.
- Project Management Excellence: Embedded PMI best practices across the IT organization, educating users, optimizing workflows, and facilitating the adoption of Project Management tools.
- Technology Strategy and Governance: Defined and maintained a regulatory framework for the evaluation, selection, and implementation of technologies, ensuring alignment with Praxair Mexico's strategic platform goals. Conducted benchmarking and evaluation of emerging technologies to identify solutions tailored to business needs.
- Multicultural Team Leadership: Led multidisciplinary teams across Mexico, USA, India, Brazil, Spain, and Argentina, fostering effective collaboration and driving successful project execution in diverse environments.

Project Leader, Praxair Mexico
Dec 1999 – Jul 2008

Key Achievements and Responsibilities:

- Business Intelligence Development: Implemented BI solutions utilizing MS SQL Server Integration Services, Business Objects, and custom applications for Accounting, Logistics, and Sales data marts. Delivered actionable insights through integrated reporting solutions (e.g., Crystal Reports and homemade database solutions).
- IT Project Management: Directed analysis, design, and implementation of IT projects, ensuring alignment with business requirements and delivering solutions within defined timeframes and scopes.
- Technology Evaluation: Conducted detailed assessments of new technologies and tools, advocating for their use and adoption where applicable to add value to business operations.
- Software Development: Designed and implemented software solutions based on Microsoft (.Net technologies), responding to user needs and enabling operational efficiencies.

Systems Administrator, Bancomer Bank
1997 - 1999.

Key Achievements and Responsibilities:

- Corporate Server Administration: Managed Windows NT and Sybase 10 / 11 servers, ensuring reliable operations for critical banking systems.
- Application Development: Designed and maintained applications in C++ and Power Builder, correcting bugs, extending functionality, and developing new features aligned with the organization's business needs.

Project Leader, Siskon Systems
1995 - 1996.

Key Achievements and Responsibilities:

- Project Design and Implementation: Directed project management for accounting, human resource administration, and risk management applications for several clients, delivering tailored solutions across diverse industries.

Certifications

Certified Information Security Manager (CISM)
October 2018

Project Management Professional (PMP) - 2080872
Sep 2017

Certified Information Systems Auditor (CISA)
Aug 2017

Education

Postgraduate Diploma in Machine Learning and Artificial Intelligence
Columbia Engineering, Columbia University, New York — March 2019 to December 2019. Certificate completed.

Machine Learning Engineer Nanodegree / Online Coursework
Udacity — March 2018 to June 2018.

Post Graduate Diploma on Project Management (PMI),
06-2008 / 11-2008. Instituto Tecnológico Autónomo de México. Certified.

Post Graduate Diploma on Development of Management Skills,
10-2003 / 02-2004. Instituto Tecnológico de Estudios Superiores de México. Certified.

Bachelor's Degree in Computer Science
Instituto Politécnico Nacional — September 1992 to August 1996. Degree completed. Certified.
Official average: 8.5 / 10.0; U.S. equivalent GPA: 3.51 / 4.0, as evaluated by Spantran (The Evaluation Company).

Languages

Spanish

English

Portuguese (Basic level)